

Problem D. Salary Queries

Time Limit 1000 ms

Mem Limit 524288 kB

A company has n employees with certain salaries. Your task is to keep track of the salaries and process queries.

Input

The first input line contains two integers n and q : the number of employees and queries.

The employees are numbered $1, 2, \dots, n$.

The next line has n integers $p_1, p_2, \dots, p_n$: each employee's salary.

After this, there are q lines describing the queries. Each line has one of the following forms:

- **!** $k\ x$: change the salary of employee k to x
- **?** $a\ b$: count the number of employees whose salary is between $a \dots b$

Output

Print the answer to each **?** query.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq p_i \leq 10^9$
- $1 \leq k \leq n$
- $1 \leq x \leq 10^9$
- $1 \leq a \leq b \leq 10^9$

Example

Input	Output
5 3 3 7 2 2 5 ? 2 3 ! 3 6 ? 2 3	3 2